



BIOINFORMATICS & DATA SCIENCE | SESSION 2 OF 2

How a computer learns

Machine learning, a real case study, and life after high school

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Three hours, four quizzes, two challenges

- 8:00 Warm-up quiz
- 8:10 How models learn
- 8:40 Challenge C: you are the model
- 9:05 Live demo: the computer finds the rule
- 9:15 Measuring a model, AlphaFold
- 9:25 Break
- 9:40 Case study: food safety
- 10:05 Challenge D: pick the threshold
- 10:30 After high school, Esame di Stato
- 10:45 Q&A

Answer live on Wooclap

Un PC per squadra, niente telefoni

01

Open

wooclap.com on your team PC

02

Join

Enter the event code

03

Vote

One answer per team, 30 seconds

EVENT CODE

XXXXXXXX

app.wooclap.com

Recap: in a FASTA file, what is on the line starting with ">"?

A

The sequence

B

The score

C

Name and description

D

The date



Name and description

That is the header. The sequence starts on the next line.

SECTION

01

How models learn

Come impara un modello



Who writes the rules?

Chi scrive le regole?

TRADITIONAL PROGRAMMING

rules + data



answers

A person writes the rules

MACHINE LEARNING

data + answers



rules

The computer learns the rules

Supervised vs unsupervised

Con o senza la risposta negli esempi

Supervised

Learns to predict an answer

e.g. Will this batch exceed the legal limit?

Unsupervised

Finds groups and patterns

e.g. Which patients are similar to each other?

Features and labels

Le colonne che descrivono e quella da prevedere

Food	Origin	Season	Non-compliant?
Maize	non-EU	summer	yes
Rice	EU	winter	no
Swordfish	EU	winter	yes
Apple	non-EU	summer	no

FEATURES

What we know
beforehand

LABEL

What we want to predict

Made-up data for teaching purposes

Judge a model on data it has never seen

Si giudica su dati mai visti

Training set 80%

Test set 20%

OVERFITTING

Like memorising the answer key: 10/10 on known exercises, 5/10 on new ones.

100% accuracy on training data, 60% on the test set. What happened?

A

Overfitting

B

The test was too easy

C

The model is perfect

D

It needs less data

A

Overfitting

It memorised the examples instead of learning a rule. Fixes: simpler models, more data, testing on unseen data.

Data leakage: peeking into the future

Il modello sbircia la risposta

Column "sample re-tested"

Predicts everything perfectly. But a sample is re-tested only after a problem is found.

At prediction time, do I really have this information?

You are the model

Sfida C, scheda C. Il modello siete voi

01

Train

12 batches. Write one rule:
"non-compliant if..."

02

Predict

Apply it to 6 new batches

03

Evaluate

How many out of 6 did you get right?

One column is empty in the new batches. Why?

One good rule, two lessons

Una buona regola, due lezioni

RULE

Non-EU maize or swordfish

4 / 6

on the test set

1 "Re-tested" was perfect in training and useless on the test.

2 On new data, some errors always remain.

Now let the computer find the rule

Ora la regola la cerca il computer

PYTHON

```
cols = ["food", "origin", "season", "retested"]
X = pd.get_dummies(train[cols])
y = train["non_compliant"]

model = DecisionTreeClassifier(random_state=0)
model.fit(X, y)

names = list(X.columns)
print(export_text(model, feature_names=names))
print("Training accuracy:", model.score(X, y))
```

OUTPUT

```
|--- retested_yes <= 0.50
|   |--- class: no
|--- retested_yes > 0.50
|   |--- class: yes

Training accuracy: 1.0
```

100% on training. Too good to be true: it found the leak, just like some of you.

Remove the leak, train again

Tolgo la colonna che sbircia il futuro

PYTHON

```
cols = ["food", "origin", "season"]
X = pd.get_dummies(train[cols])

model = DecisionTreeClassifier(random_state=0)
model.fit(X, y)

names = list(X.columns)
print(export_text(model, feature_names=names))
```

OUTPUT

```
|--- food_Swordfish <= 0.50
|   |--- food_Maize <= 0.50
|   |   |--- class: no
|   |--- food_Maize > 0.50
|   |   |--- origin_EU <= 0.50
|   |   |   |--- class: yes
|   |   |--- origin_EU > 0.50
|   |   |   |--- class: no
|--- food_Swordfish > 0.50
|   |--- class: yes
```

Swordfish, or non-EU maize. The same rule you found by hand.

Six new batches: the machine scores like you

Sei lotti nuovi: il computer fa come voi

Batch	Food	Origin	Season	Prediction	Actual
T1	Maize	non-EU	summer	yes	yes
T2	Swordfish	EU	summer	yes	yes
T3	Rice	EU	summer	no	no
T4	Maize	EU	summer	no	yes
T5	Apple	non-EU	summer	no	no
T6	Maize	non-EU	winter	yes	no

4 / 6

correct, like the rule you
found

Faster than you, not smarter. A model is only as good as its data.

When 95% means nothing

Quando l'accuratezza inganna

95%

accuracy of a model that always says "compliant"

0

problems found

Recall

Of the risky batches, how many do I catch?

Precision

When I say "risky", how often am I right?

100 batches, 5 risky. The model flags 10, including 4 of the 5. What is its recall?

A

40%

B

50%

C

80%

D

96%

C

80%: it catches 4 of 5

Precision is 4 out of 10, i.e. 40%. Two different questions: improving one usually hurts the other.

From sequence to shape

Dalla sequenza alla forma 3D

50 years

an open problem

200M+

predicted structures, free to use

Nobel

Chemistry, 2024

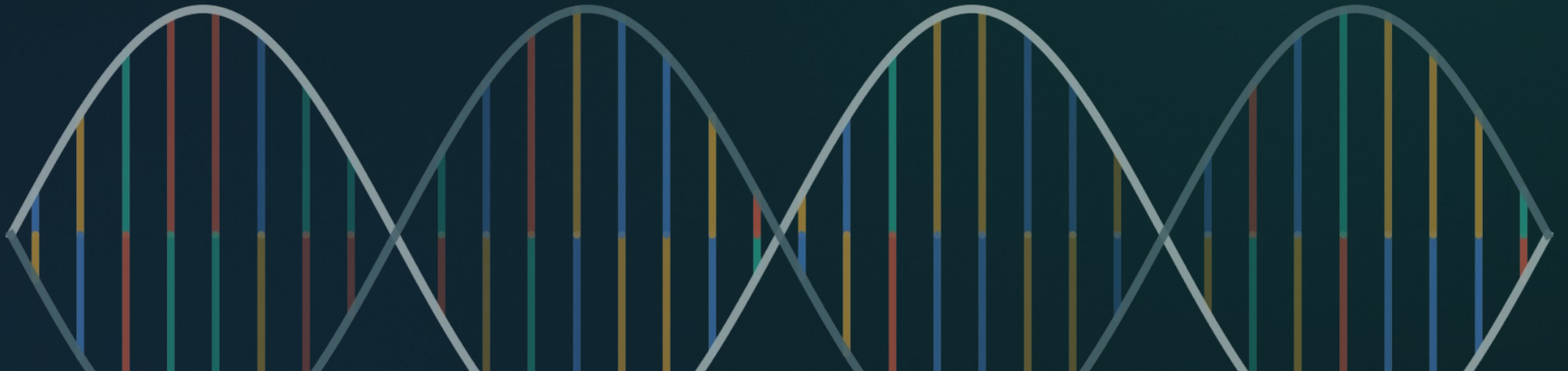
A sequence goes in. A 3D protein comes out.



alphafold.ebi.ac.uk ↗

Break

15 minutes. Then a real case.



SECTION

02

A real case study

Il mio tirocinio a Neutron Laboratories



Where should the lab look harder?

Dove conviene controllare di più?



Mycotoxins

from moulds, e.g. in maize



Plant toxins

made by plants



Metals

lead, cadmium, mercury



Pollutants

dioxins and PCBs



Process

formed during processing



Nitrates

leafy vegetables

The 6 contaminant families in EU Regulation 2023/915

The data

I dati del laboratorio

182,033

lab analyses

122

contaminants

6

regulation sections

Goal for each contaminant and food pair, the probability of exceeding the legal limit.

From database to decision

Dal database alla decisione

01

Extract

SQL on the lab
database

02

Clean

Formats and
units

03

Analyse

Statistics

04

Predict

CatBoost

05

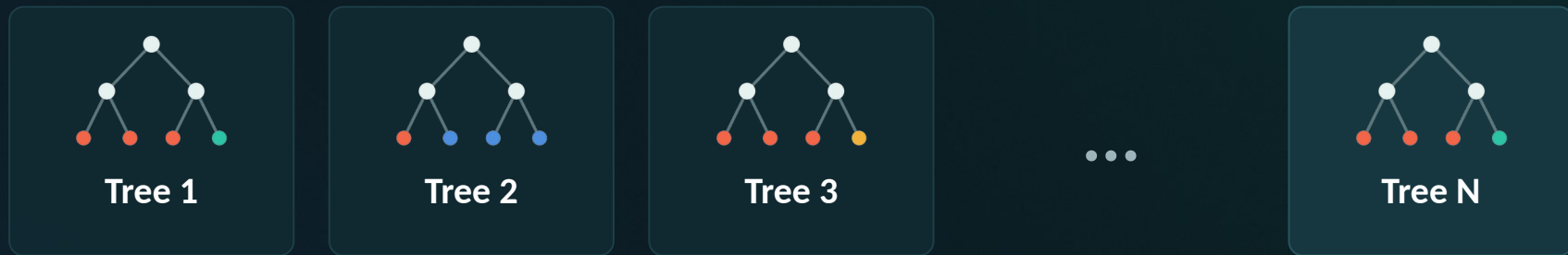
Decide

High, medium,
low risk

Then the real test: months the model had never seen.

Many small trees

Gradient boosting con CatBoost



Each tree fixes the mistakes of the ones before.

The final prediction is the sum of all of them.

Results

I risultati

	Predicted risky	Predicted OK
Actually risky	50	7
Actually OK	31	216

87.7%

recall: 50 of 57 caught

21 / 22

caught on months never seen

Contaminant and food pairs in the test set, threshold 0.50

In food safety, which error is usually worse?

A

They are equally bad

B

False negative: a risky batch gets through

C

False positive: a safe batch is re-tested

D

Only the price of the test matters

A blue square with rounded corners containing a large, bold, black letter 'B'.

The false negative

A false positive costs one extra test. A false negative can put a non-compliant product on the market. But checks are not unlimited.

Pick the threshold

Sfida D, scheda D. Scegliete la soglia

20

batches with a model
probability

€200

per check

€5,000

per missed risky batch

Round 2 the lab can run at most 10 checks. Which threshold do you choose?

No threshold is right for every case

Nessuna soglia giusta in assoluto

Threshold	Checks	Missed	Cost
0.3	13	1	€7,600
0.5	8	2	€11,600
0.7	4	4	€20,800
0.4	10	1	€7,000

← with the 10-check limit

It depends on costs and constraints.

SECTION

03

After high school

Percorsi e lavoro



Where people work with biological data

Dove si lavora con i dati biologici



Research

universities, EMBL-EBI



Health

genomics, digital health



Pharma & biotech

drug discovery



Food & environment

labs, One Health



Tech

data, dashboards, software



Science communication

schools, publishing

From your computer science class to a hospital

Dalle query di informatica ai dati di un ospedale

SQL

```
SELECT hospital,  
       COUNT(*) AS reports,  
       SUM(signed) AS signed  
FROM reports  
GROUP BY hospital;
```

RESULT

hospital	reports	signed
A	100	90
B	900	450

A: 90% B: 50%

A signs 90%, B signs 50%. Together, do they sign 70%?

Many roads, one destination

Più strade portano allo stesso punto

01

High school

maths, biology,
English

02

Bachelor's

biology, biotech,
computer science,
maths, physics

03

Master's

Molecular and
Computational
Biology, Bologna

04

Then

industry, PhD, abroad

Up-to-date requirements: corsi.unibo.it ↗

How to get in

Come si entra: triennali a Bologna e test TOLC

Bachelor's in Bologna	Language	Places 2026/27	Test
<u>Biological sciences</u>	Italian	200	TOLC-I
<u>Biotechnology</u>	Italian	125	TOLC-I
<u>Computer science</u>	Italian	150	TOLC-I
<u>Genomics</u>	English	limited	own call

4th & 5th year

you can already sit the TOLC

Once a month

at most, so you can retry

cisiaonline.it ↗

where you book it

Places and rules change every year: check the official call (bando) on corsi.unibo.it ↗

Click a degree name to open its admission page.

What you can start today

Cosa potete fare già adesso



Python

a few weeks is enough to start



rosalind.info

free bioinformatics problems



English

almost everything is in English



Statistics

to tell a real result from chance

python.org ↗

rosalind.info ↗

Ideas for your oral exam

Spunti per il colloquio: collegamenti tra materie

DNA as information

Biology: genetic code, mutations

Maths: 4^3 codons, combinatorics

Computer science: binary encoding

Sickle cell and malaria

Biology: point mutation, natural selection

Geography: where malaria spreads

Statistics: frequency of a variant

Food safety, One Health

Chemistry: contaminants, legal limits

Civic education: EU rules, consumer health

Statistics: risk and errors

AI in science

Chemistry: AlphaFold, Nobel 2024

Philosophy: can a machine learn?

Computer science: training and testing

New: digital health CS: databases and SQL · Civics: health records and privacy · Maths: weighted average

All the code and quizzes, online

Tutto il codice e i quiz, da rifare a casa



Code

every live demo, ready for Google Colab

Quiz

all the questions, with the explanation

Links

BLAST, UniProt, AlphaFold, Rosalind, Unibo

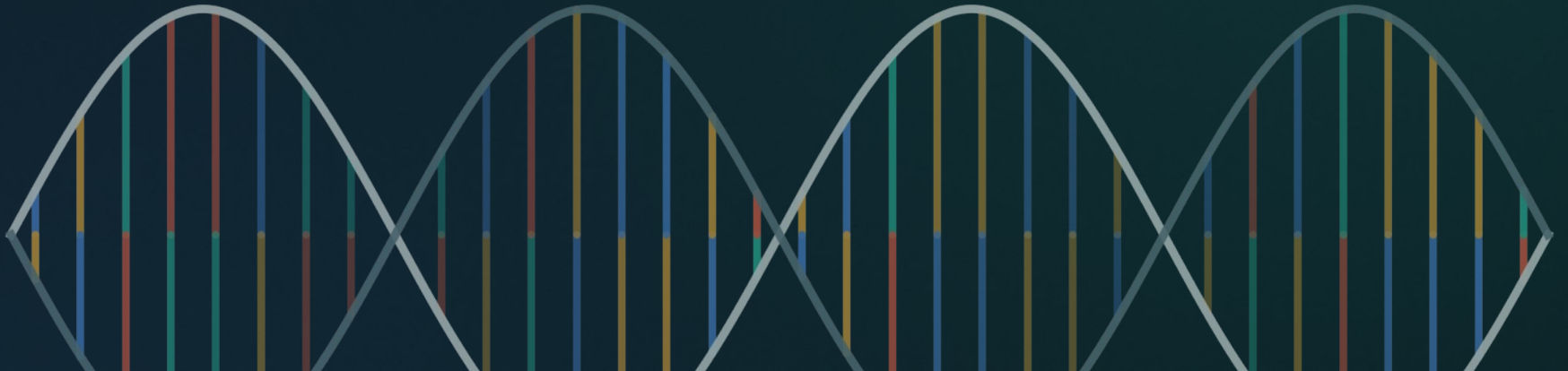
Feedback

anonymous, 2 min: link from your teachers

martinaa1408.github.io/liceo-ferrari-bioinformatics ↗

Questions?

University, jobs, anything from today.





Thank you

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